

Weak A2 spaces, the Kastanas game and strategically Ramsey sets

Clement Yung, University of Toronto

19 Aug 2026



Example.

1. X is infinite.
2. Structure = Infinite set.

Fact (Pigeonhole principle)

Let X be an infinite set. If $Y_0, \dots, Y_{n-1}$ is a partition of X , then there exists some $i < n$ such that Y_i is infinite.

Example. Let $\mathbb{F}$ be a countable (possibly finite) field. Let E be a vector space over $\mathbb{F}$ of dimension $\aleph_0$, with Hamel basis $(e_n)_{n=0}^\infty$.

Definition

An *infinite block sequence* is a $<$ -increasing sequence of non-zero vectors of E .

Now consider the following setting:

1. $X = E \setminus \{0\}$, the set of non-zero vectors.
2. Structure = Infinite-dimensional block subspace (without 0).

Does the Ramsey theorem hold for this variant?

This theorem fails if $|\mathbb{F}| > 2$. We define the set Y as:

$$Y = \{x \in E \setminus \{0\} : x = e_n + y \text{ for some } e_n < y\},$$

$$= \left\{ x \in E \setminus \{0\} : \begin{array}{l} x = e_{n_0} + \lambda_{n_1} e_{n_1} + \cdots + \lambda_{n_k} e_{n_k} \\ \text{and } n_0 < \cdots < n_k \end{array} \right\}.$$

Then Y, Y^c partition $E \setminus \{0\}$, but neither Y nor Y^c contains $V \setminus \{0\}$ for any infinite-dimensional block subspace V .

Example

Let $A = (e_0 + e_1, e_2 + e_3, \dots)$. Then $e_0 + e_1 \in Y$, but $2e_0 + 2e_1 \in Y^c$, so $\text{span}(A) \setminus \{0\}$ is not a subset of either Y or Y^c .

Infinite-dimensional Ramsey theory

For infinite-dimensional Ramsey theory, we partition $X^{\mathbb{N}}$ or a closed subset $\mathcal{R}$ of $X^{\mathbb{N}}$. Here we equip X with the discrete topology, and $X^{\mathbb{N}}$ with the product topology.

More precisely:

Let X be a set or structure. Let $\mathcal{X}_0, \dots, \mathcal{X}_{n-1}$ be a partition of $\mathcal{R}$, a closed subset of $X^{\mathbb{N}}$. Does there exist some $i < n$ such that $\mathcal{X}_i$ contains some substructure of interest?

Let $[\mathbb{N}]^{<\infty} \restriction A := \{a \in [\mathbb{N}]^{<\infty} : a \subseteq A\}$.

Notation

Given $A \in [\mathbb{N}]^\infty$ and $a \in [\mathbb{N}]^{<\infty} \upharpoonright A$, we write:

$$[a, A] := \{B \in [\mathbb{N}]^\infty : a \sqsubseteq B \text{ and } B \subseteq A\}$$

where $a \sqsubseteq B$ means that $B \cap (\max(a) + 1) = a$.

Each $[a, A]$ is also called an *Ellentuck neighbourhood*.

Example. Let E be a vector space over a countable field $\mathbb{F}$ of dimension $\aleph_0$, with Hamel basis $(e_n)_{n=0}^\infty$.

1. $X = E$.
2. $\mathcal{R} = E^{[\infty]}$, the set of all infinite block sequences of vectors ($\Leftrightarrow$ infinite-dimensional block subspaces).
3. Structure = Ellentuck neighbourhood of infinite-dimensional block subspaces.

Definition

A set $\mathcal{X} \subseteq E^{[\infty]}$ is *Ramsey* if for all $A \in E^{[\infty]}$ and $a \in E^{[<\infty]} \upharpoonright A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Definition

A set $\mathcal{X} \subseteq E^{[\infty]}$ is *Ramsey* if for all $A \in E^{[\infty]}$ and $a \in E^{[<\infty]} \upharpoonright A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Fact

Assuming the Axiom of Choice, there is a subset $\mathcal{X} \subseteq E^{[\infty]}$ that is not Ramsey.

We observe some patterns:

We observe some patterns:

1. $\mathcal{R}$ is a set of infinite increasing sequences under some partial order $<$.

We observe some patterns:

1. $\mathcal{R}$ is a set of infinite increasing sequences under some partial order $<$.
2. Using either $\subseteq$ or $\leq$, we defined the Ellentuck neighbourhood $[a, A]$, and the notion of a Ramsey subset $\mathcal{X} \subseteq \mathcal{R}$.

We observe some patterns:

1. $\mathcal{R}$ is a set of infinite increasing sequences under some partial order $<$.
2. Using either $\subseteq$ or $\leq$, we defined the Ellentuck neighbourhood $[a, A]$, and the notion of a Ramsey subset $\mathcal{X} \subseteq \mathcal{R}$.

Question. Can this pattern be captured and made into an abstract framework?

Notation

Given a triple $(\mathcal{R}, \leq, r)$, for $A \in \mathcal{R}$ and $a \in \mathcal{AR}$:

$$a \sqsubseteq A \iff a = r_n(A) \text{ for some } n \in \mathbb{N}.$$

Notation

If $A \in \mathcal{R}$ and $a \in \mathcal{AR}$,

$$[a, A] := \{B \in \mathcal{R} : a \sqsubseteq B \text{ and } B \leq A\}.$$

Axiom (**A1**, Sequencing)

- (1) $r_0(A) = \emptyset$ for all $A \in \mathcal{R}$.
- (2) $A \neq B$ implies $r_n(A) \neq r_n(B)$ for some n .
- (3) $r_n(A) = r_m(B)$ implies $n = m$ and $r_k(A) = r_k(B)$ for all $k < n$.

Axiom (**A2**, Finitisation)

There is a quasi-ordering $\leq_{\text{fin}}$ on $\mathcal{AR}$ such that:

- (1) $\{b \in \mathcal{AR} : b \leq_{\text{fin}} a\}$ is **finite** for all $a \in \mathcal{AR}$.
- (2) $A \leq B$ iff $\forall n \exists m [r_n(A) \leq_{\text{fin}} r_m(B)]$.
- (3) $\forall a, b \in \mathcal{AR} [a \sqsubseteq b \wedge b \leq_{\text{fin}} c \rightarrow \exists d \sqsubseteq c [a \leq_{\text{fin}} d]]$.

1. $([\mathbb{N}]^\infty, \subseteq, r)$ satisfies **A2**, as for all $a \in [\mathbb{N}]^{<\infty}$, $\{b : b \subseteq a\}$ is finite.

Axiom (**A2**, Finitisation)

There is a quasi-ordering $\leq_{\text{fin}}$ on $\mathcal{AR}$ such that:

- (1) $\{b \in \mathcal{AR} : b \leq_{\text{fin}} a\}$ is **finite** for all $a \in \mathcal{AR}$.
- (2) $A \leq B$ iff $\forall n \exists m [r_n(A) \leq_{\text{fin}} r_m(B)]$.
- (3) $\forall a, b \in \mathcal{AR} [a \sqsubseteq b \wedge b \leq_{\text{fin}} c \rightarrow \exists d \sqsubseteq c [a \leq_{\text{fin}} d]]$.

1. $([\mathbb{N}]^\infty, \subseteq, r)$ satisfies **A2**, as for all $a \in [\mathbb{N}]^{<\infty}$, $\{b : b \subseteq a\}$ is finite.
2. $(E^{[\infty]}, \leq, r)$ satisfies **A2** if $|\mathbb{F}| < \infty$, as if $a = (x_i)_{i < n} \in E^{[<\infty]}$, then there are finitely many subspaces of a .

Axiom (**A2**, Finitisation)

There is a quasi-ordering $\leq_{\text{fin}}$ on $\mathcal{AR}$ such that:

- (1) $\{b \in \mathcal{AR} : b \leq_{\text{fin}} a\}$ is **finite** for all $a \in \mathcal{AR}$.
- (2) $A \leq B$ iff $\forall n \exists m [r_n(A) \leq_{\text{fin}} r_m(B)]$.
- (3) $\forall a, b \in \mathcal{AR} [a \sqsubseteq b \wedge b \leq_{\text{fin}} c \rightarrow \exists d \sqsubseteq c [a \leq_{\text{fin}} d]]$.

1. $([\mathbb{N}]^\infty, \subseteq, r)$ satisfies **A2**, as for all $a \in [\mathbb{N}]^{<\infty}$, $\{b : b \subseteq a\}$ is finite.
2. $(E^{[\infty]}, \leq, r)$ satisfies **A2** if $|\mathbb{F}| < \infty$, as if $a = (x_i)_{i < n} \in E^{[<\infty]}$, then there are finitely many subspaces of a .
3. $(E^{[\infty]}, \leq, r)$ does not satisfy **A2** if $|\mathbb{F}| = \infty$, as if $a = (x_0, x_1)$, then $\text{span}\{\lambda x_0 + x_1\}$ is a (block) subspace for any $\lambda \in \mathbb{F}$.

We let $\mathcal{AR}_n$ be the image of the map $r_n(-)$, i.e. the set of all finite approximations of length n .

Axiom (A4, Pigeonhole)

If $\text{depth}_B(a) < \infty$ and if $\mathcal{O} \subseteq \mathcal{AR}_{\text{lh}(a)+1}$, then there exists $A \in [\text{depth}_B(a), B]$ such that $r_{\text{lh}(a)+1}[a, A] \subseteq \mathcal{O}$ or $r_{\text{lh}(a)+1}[a, A] \subseteq \mathcal{O}^c$.

Pigeonhole should be viewed as a statement about the finite-dimensional Ramsey theorem for $\mathcal{R}$.

1. $([\mathbb{N}]^\infty, \subseteq, r)$ satisfies **A4**, due to the pigeonhole principle.

We let $\mathcal{AR}_n$ be the image of the map $r_n(-)$, i.e. the set of all finite approximations of length n .

Axiom (**A4**, Pigeonhole)

If $\text{depth}_B(a) < \infty$ and if $\mathcal{O} \subseteq \mathcal{AR}_{\text{lh}(a)+1}$, then there exists $A \in [\text{depth}_B(a), B]$ such that $r_{\text{lh}(a)+1}[a, A] \subseteq \mathcal{O}$ or $r_{\text{lh}(a)+1}[a, A] \subseteq \mathcal{O}^c$.

Pigeonhole should be viewed as a statement about the finite-dimensional Ramsey theorem for $\mathcal{R}$.

1. $([\mathbb{N}]^\infty, \subseteq, r)$ satisfies **A4**, due to the pigeonhole principle.
2. $(E^{[\infty]}, \leq, r)$ satisfies **A4** if $|\mathbb{F}| = 2$, due to Hindman's theorem.
3. $(E^{[\infty]}, \leq, r)$ does not satisfy **A4** if $|\mathbb{F}| > 2$, as Hindman's theorem fails for $|\mathbb{F}| > 2$.

Observe that every $A \in \mathcal{R}$ can be uniquely identified with the sequence $(r_n(A))_{n=0}^\infty$, an element of $\mathcal{AR}^\mathbb{N}$. Therefore, we may identify $\mathcal{R}$ with a subset of $\mathcal{AR}^\mathbb{N}$.

Recall that in infinite-dimensional Ramsey theory, we require $\mathcal{R}$ to be a closed subset of $X^{\mathbb{N}}$. We make a similar requirement here.

Definition

$(\mathcal{R}, \leq, r)$ is a *closed triple* if for every $\sqsubseteq$ -increasing sequence $(a_n)_{n=0}^\infty$ of elements in $\mathcal{AR}$ such that $\text{lh}(a_n) = n$, there exists some $A \in \mathcal{R}$ such that $r_n(A) = a_n$ for all n .

In other words, $\mathcal{R}$ is a metrically closed subset of $\mathcal{AR}^{\mathbb{N}}$.

Definition

A *topological Ramsey space* is a closed triple $(\mathcal{R}, \leq, r)$ satisfying **A1–A4**.

Definition

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space. A set $\mathcal{X} \subseteq \mathcal{R}$ is *Ramsey* if for all $A \in \mathcal{R}$ and $a \in \mathcal{AR} \upharpoonright A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Definition

A *topological Ramsey space* is a closed triple $(\mathcal{R}, \leq, r)$ satisfying **A1–A4**.

Definition

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space. A set $\mathcal{X} \subseteq \mathcal{R}$ is *Ramsey* if for all $A \in \mathcal{R}$ and $a \in \mathcal{AR} \restriction A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Theorem (Todorćević, 2010)

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space such that $\mathcal{AR}$ is countable. Then every analytic set is Ramsey.

Examples

$(\mathcal{R}, \leq, r)$	Closed triple	A1	A2	A3	A4	Topological Ramsey space
$([\mathbb{N}]^\infty, \subseteq, r)$	✓	✓	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = 2$	✓	✓	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $2 < \mathbb{F} < \infty$	✓	✓	✓	✓		
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = \infty$	✓	✓		✓		

Banach space theory

It turns out that there is a deep connection between Banach space theory and the Ramsey theory for (countable) vector spaces.

Banach space theory

It turns out that there is a deep connection between Banach space theory and the Ramsey theory for (countable) vector spaces.

Question (Banach, 1932)

Let X be an infinite-dimensional Banach space, and suppose that X is *homogeneous* (i.e. X is isomorphic to every one of its infinite-dimensional closed subspaces). Must X be isomorphic to ℓ^2 ?

Banach space theory

It turns out that there is a deep connection between Banach space theory and the Ramsey theory for (countable) vector spaces.

Question (Banach, 1932)

Let X be an infinite-dimensional Banach space, and suppose that X is *homogeneous* (i.e. X is isomorphic to every one of its infinite-dimensional closed subspaces). Must X be isomorphic to ℓ^2 ?

This was answered in the positive by Gowers (2002), and the proof is heavily Ramsey-theoretic and uses ideas from infinite games.

We fix some countable vector space E . Let $A \in E^{[\infty]}$ be an infinite block sequence, and let $a \in E^{[<\infty]} \upharpoonright A$.

Definition (Gowers, 2002)

The *Gowers game* played below $[a, A]$, denoted as $G[a, A]$, is the following game:

$$\begin{array}{c|ccc}
 \text{I} & A_0 \leq A & A_1 \leq A & \dots \\
 \hline
 \text{II} & x_0 \in \langle A_0 \rangle & x_1 \in \langle A_1 \rangle & \dots
 \end{array}$$

where $\langle A_i \rangle$ is the span of the elements in the sequence A_i . The outcome of this game is the sequence $a \frown (x_k)_{k=0}^\infty \in E^{[\infty]}$.

Definition

We say that **I** (similarly **II**) has a strategy in a game to *reach* $\mathcal{X} \subseteq E^{[\infty]}$ if they have a strategy in the game to ensure the outcome is in $\mathcal{X}$.

Definition (Rosendal, 2010)

A subset $\mathcal{X} \subseteq E^{[\infty]}$ is *strategically Ramsey* if for all $A \in E^{[\infty]}$ and $a \in E^{[<\infty]} \upharpoonright A$, there exists some $B \in [a, A]$ such that one of the following holds:

1. I has a strategy in $F[a, B]$ to reach $\mathcal{X}^c$.
2. II has a strategy in $G[a, B]$ to reach $\mathcal{X}$.

Theorem (Rosendal, 2010)

Every analytic subset of $E^{[\infty]}$ is strategically Ramsey.

Note that there is a coanalytic non-strategically Ramsey set in Gödel's constructible universe L .

wA2-spaces and the Kastanas game

$(\mathcal{R}, \leq, r)$	A2	A4	Topological Ramsey space	Strategically Ramsey sets
$([\mathbb{N}]^\infty, \subseteq, r)$	✓	✓	✓	
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = 2$	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $2 < \mathbb{F} < \infty$	✓			✓
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = \infty$				✓

wA2-spaces and the Kastanas game

$(\mathcal{R}, \leq, r)$	A2	A4	Topological Ramsey space	Strategically Ramsey sets
$([\mathbb{N}]^\infty, \subseteq, r)$	✓	✓	✓	
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = 2$	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $2 < \mathbb{F} < \infty$	✓			✓
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = \infty$				✓

Question. Is there a framework unifying topological Ramsey theory and the theory of strategically Ramsey sets?

We shall unify the two frameworks by doing two things.

1. Weakening the axioms **A1–A4** to include countable vector spaces with $|\mathbb{F}| > 2$.
2. Finding a “common game” that connects Ramsey sets to strategically Ramsey sets.

All examples we saw are **wA2**-spaces:

$(\mathcal{R}, \leq, r)$	Closed triple	A1	A2	wA2	A3	A4
$([\mathbb{N}]^\infty, \subseteq, r)$	✓	✓	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = 2$	✓	✓	✓	✓	✓	✓
$(E^{[\infty]}, \leq, r)$ $2 < \mathbb{F} < \infty$	✓	✓	✓	✓	✓	
$(E^{[\infty]}, \leq, r)$ $ \mathbb{F} = \infty$	✓	✓		✓	✓	

Theorem (Kastanas, 1983)

A set $\mathcal{X} \subseteq [\mathbb{N}]^\infty$ is Ramsey iff it is Kastanas Ramsey.

Theorem (Kastanas, 1983)

A set $\mathcal{X} \subseteq [\mathbb{N}]^\infty$ is Ramsey iff it is Kastanas Ramsey.

By the Borel determinacy theorem for games on reals, we get:

Corollary

Every Borel subset of $[\mathbb{N}]^\infty$ is Ramsey.

Since analytic determinacy is not provable without large cardinals, we cannot immediately conclude that every analytic subset of $[\mathbb{N}]^\infty$ is Ramsey (even though it is true).

Definition

A set $\mathcal{X} \subseteq \mathcal{R}$ is *Kastanas Ramsey* if for all $A \in \mathcal{R}$ and $a \in \mathcal{AR} \restriction A$, there exists some $B \in [a, A]$ such that one of the following holds:

1. I has a strategy in $K[a, B]$ to reach $\mathcal{X}^c$.
2. II has a strategy in $K[a, B]$ to reach $\mathcal{X}$.

Cano and Di Prisco showed that for a class of topological Ramsey spaces, a set is Ramsey iff it is Kastanas Ramsey.

Theorem (Y., 2024)

If $(\mathcal{R}, \leq, r)$ is a topological Ramsey space, then $\mathcal{X} \subseteq \mathcal{R}$ is Ramsey iff it is Kastanas Ramsey.

Again, by the Borel determinacy theorem for games on reals:

Corollary

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space such that $\mathcal{AR}$ is countable. Then every Borel subset of $\mathcal{R}$ is Ramsey.

So for $([\mathbb{N}]^\infty, \subseteq, r)$ and $(E^{[\infty]}, \leq, r)$ with $|\mathbb{F}| = 2$, Kastanas Ramsey sets are precisely the Ramsey sets. What about $(E^{[\infty]}, \leq, r)$ with $|\mathbb{F}| > 2$ (which are also **wA2**-spaces)?

So for $([\mathbb{N}]^\infty, \subseteq, r)$ and $(E^{[\infty]}, \leq, r)$ with $|\mathbb{F}| = 2$, Kastanas Ramsey sets are precisely the Ramsey sets. What about $(E^{[\infty]}, \leq, r)$ with $|\mathbb{F}| > 2$ (which are also **wA2**-spaces)?

Theorem (Y., 2024)

A subset $\mathcal{X} \subseteq E^{[\infty]}$ is Kastanas Ramsey iff it is strategically Ramsey.

Let's recall the two results about analytic sets.

Theorem (Todorčević, 2010)

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space such that $\mathcal{AR}$ is countable. Then every analytic set is Ramsey.

Theorem (Rosendal, 2010)

Every analytic subset of $E^{[\infty]}$ is strategically Ramsey.

Let's recall the two results about analytic sets.

Theorem (Todorčević, 2010)

Let $(\mathcal{R}, \leq, r)$ be a topological Ramsey space such that $\mathcal{AR}$ is countable. Then every analytic set is Ramsey.

Theorem (Rosendal, 2010)

Every analytic subset of $E^{[\infty]}$ is strategically Ramsey.

Question. Can we unify these two statements into a single statement about analytic subsets of a **wA2**-space?

Theorem (Y., 2024)

*Let $(\mathcal{R}, \leq, r)$ be a **wA2**-space such that $\mathcal{AR}$ is countable. Then every analytic subset of $\mathcal{R}$ is Kastanas Ramsey.*

We shall spend the remainder of the talk discussing the idea of the proof.

Let $(\mathcal{R}, \leq, r)$ be a **wA2**-space such that $\mathcal{AR}$ is countable. We consider the triple $(\mathcal{R} \times 2^{\mathbb{N}}, \preceq, r)$ in the following manner:

1. $(A, u) \preceq (B, v) \iff A \leq B$.
2. $r_n(A, u) = (r_n(A), u \upharpoonright n)$.

Note that $\preceq$ is a quasi-order, but it is *not* a partial order.

Lemma

$(\mathcal{R} \times 2^{\mathbb{N}}, \preceq, r)$ is a **wA2**-space that does *not* satisfy **A4**.

This is because we can partition $\mathcal{R} \times 2^{\mathbb{N}}$ into two sets depending on the first binary bit of each $(A, u) \in \mathcal{R} \times 2^{\mathbb{N}}$. That is, we define:

$$\mathcal{X} := \{(A, u) \in \mathcal{R} \times 2^{\mathbb{N}} : u(0) = 0\}.$$

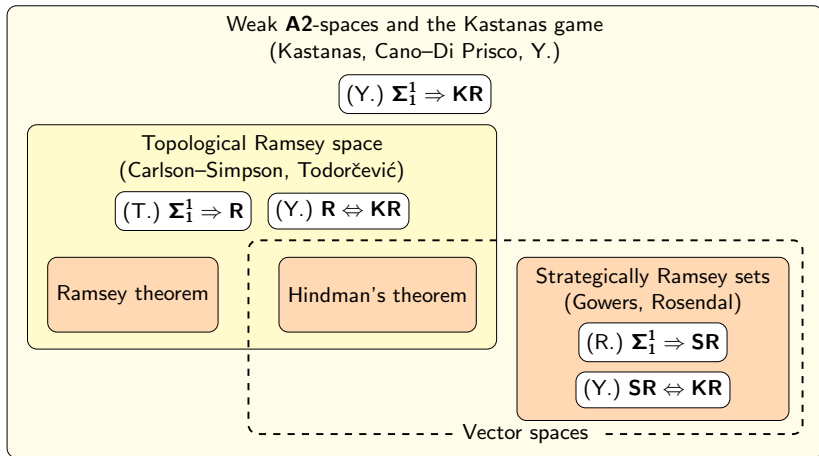
Then $\mathcal{X}$ is a clopen non-Ramsey subset of $\mathcal{R} \times 2^{\mathbb{N}}$.

Let $\pi_0 : \mathcal{R} \times 2^{\mathbb{N}} \rightarrow \mathcal{R}$ be the projection map onto the first coordinate (i.e. $\pi_0(A, u) = A$ for all $(A, u) \in \mathcal{R} \times 2^{\mathbb{N}}$).

Theorem (Y., 2024)

If $C \subseteq \mathcal{R} \times 2^{\mathbb{N}}$ is Kastanas Ramsey, then $\pi_0''(C) \subseteq \mathcal{R}$ is also Kastanas Ramsey.

Thanks for listening!



R = Ramsey, KR = Kastanas Ramsey, SR = Strategically Ramsey, Σ_1^1 = Analytic.